Comparing models

Was this page helpful?Â

Whyâ

At each iteration of your model, you will need to evaluate it. In reality, not only evaluate it but also assess how it performs in comparison with the previous iteration.

Pulse helps you in taking your conclusions on how your feature engineering work, model training, and parameter tuning is evolving and progressing towards the goals of your Data Science project.

It may make sense to, for example, try out your model not only with adjusted parameters but also for different test datasets, different transaction time periods, etc.

How toâ

While you evaluate your models, Pulse enables you to compare their performances:

1. Directly by overlaying model evaluations.
2. Using the exploration mode for business targets.

Model Evaluationsâ

To compare two **Model Evaluations**, execute the following actions:

1. In the **Data Science** tab, select a **Model Project** from the **Model Projects** list.
2. Select an existing **Model Evaluation**.
3. Select which is the evaluation you want the current one to be **Compared With**.

The information in **Model Evaluation** is divided into different panels that can be accessed for: Comparing **Performance Metrics** or Comparing **Feature importance**.

Comparing Performance Metricsâ

1. To visualize comparison results:

    1. Analyze the **ROC Curve** and **X/Y Plot**, now with both the **Model Evaluation** and the **Comparison** evaluation visible.

    2. Analyze the Comparison results displayed at the top row of the **Metrics** table.

Comparison results

When you select a **Model Evaluation** for comparison, the **ROC Curve**, **X/Y Plot** and **Metrics** table display the evaluation results only for the main combination of breakdowns (All x All). This occurs even if the model evaluation has configured breakdowns.

2. Select the Comparison Details option to display evaluation parameters for both the selected and the comparison models. Differences are highlighted so that you can easily spot all parameter adjustments you are testing.

Comparing Feature Importanceâ

This option is only available if the current Model Evaluation has Feature Importance computed (i.e. regardless of whether or not the model evaluation under comparison has it computed).

1. Select to visualize comparison results:

The following metrics are presented by default:

- **#FPR**: Values of the count FPR for each feature for the current **Model Evaluation** (A) and for the evaluation under comparison (B).
- **Absolute importance**: Bar chart comparing the **Absolute Importance** of both model evaluations.
- **Rank**: The rank of the importance of each feature for each model evaluation.
- **Delta**: The difference between Rank A and Rank B.

Feature Importance view

If you compare the current model evaluation with an evaluation that does not have feature importance computed, the **Feature Importance** view will only display data for the current model evaluation.

Exploration Modeâ

To compare how different **Model Evaluations** perform for a Target, execute the following actions:

1. In the **Data Science** tab, select a **Model Project** from the **Model Projects list**.
2. Select the Exploration Mode option to explore evaluation results in your **Model Project**.
3. [Configure Targets for evaluation](#).
4. Analyze the results for each **Target** and compare them among all selected **Evaluations**.

Comparing models

You can compare models using the exploration mode within a hyperparameter tuning or using the global exploration mode in your model project, where you can combine results from model evaluations and hyperparameter tuning.

Next Stepsâ

If your model produces the desired results, then your project is ready for deployment into Production. Check also how Pulse projects work for some background to understand what is involved at this stage.

If your model needs further tuning, then start a new iteration in your Data Science project. For example, by improving your model's features and adjusting parameters.

To add an extra layer of security on top of your Machine Learning model, you can also build and evaluate rules.